Practice workbook · Explain where bile comes from and how it aids fat digestion

Bile production in the liver, storage in the gall bladder, emulsification of fat and neutralising stomach acid, including liver damage and bile-free tubes.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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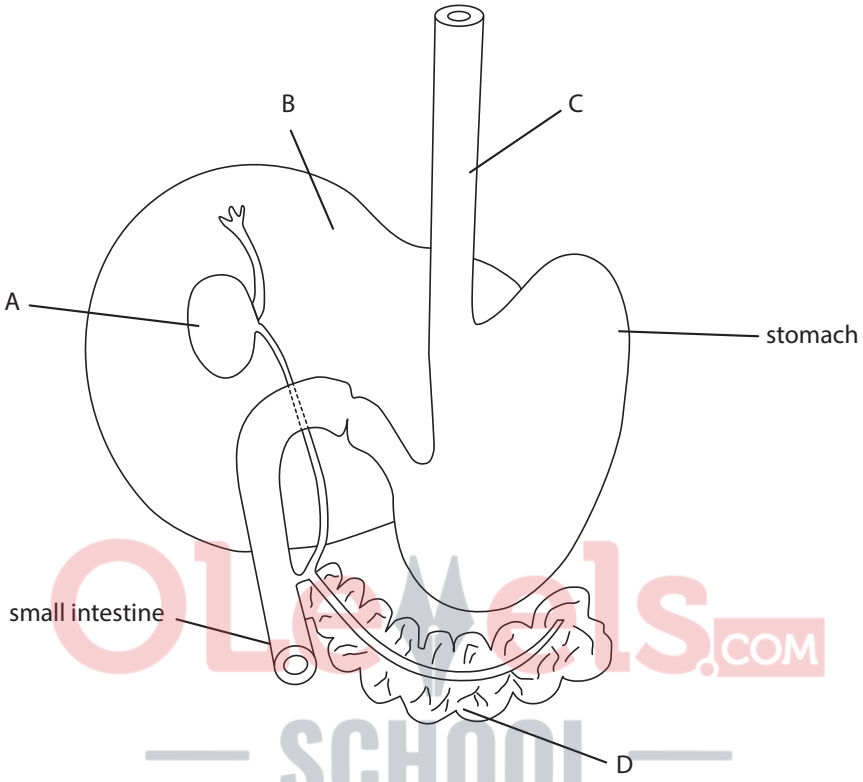
Type ID: HBI-BILE-EMULSIFY-NEUTRALISE

1. Question 2 · part a.iii · 3 marks

June 2025 Paper 2

Worked-example retry · Attempt part a.iii only.

2 The diagram shows structures involved in digestion.



(a) (i) Identify the structures labelled A, B, C and D.

(4)

A .....

B .....

C .....

D .....

(ii) Name the fluid stored in structure A.

(1)

.....

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P 7 8 7 6 9 A 0 4 2 0

1. Question 2 · part a.iii · 3 marks

June 2025 Paper 2

Worked-example retry · Attempt part a.iii only.

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(iii) Explain the function of the fluid stored in structure A in digestion.

(3)

(b) (i) Describe the role of structure D in digestion.

(4)

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(ii) Name two substances secreted by structure D that are not involved in digestion.

(2)

1 .....

2 .....

(Total for Question 2 = 14 marks)

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1. Question 2 · part a.iii · 3 marks · Working and answer

Working and answer

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2. Question 7 · part c · 5 marks

June 2019 Paper 2

Further practice · Attempt part c only.

7 The volume of alcohol in alcoholic drinks can be measured in units.

One unit is equivalent to 10 cm^3 of pure alcohol.

The diagram shows a bottle of wine with 14% alcohol content and a glass containing 250 cm^3 of this wine.



250 cm^3 glass of Ruby Red wine

(a) Calculate the number of units of alcohol in the glass of wine.

(2)

number of units = ...

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2. Question 7 · part c · 5 marks

June 2019 Paper 2

Further practice · Attempt part c only.

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(b) It is illegal for a person to drive a motor vehicle if they have more than the legal limit of alcohol in their bloodstream.

It takes one hour for the body to process one unit of alcohol so that there is no alcohol left in the bloodstream.

(i) A person drinks three units of alcohol. This puts her at the legal limit.

Calculate the time it would take before a person can legally drive after drinking four glasses of Ruby Red wine.

(2)

time = ... hours

(ii) Suggest why it is illegal for people to drive a motor vehicle with more than the legal limit of alcohol in their bloodstream.

(2)

.....

.....

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2. Question 7 · part c · 5 marks

June 2019 Paper 2

Further practice · Attempt part c only.

(c) Drinking a lot of alcohol over a long period of time can cause a condition called cirrhosis.

This condition prevents the cells of the liver from functioning normally.

Explain the effect of cirrhosis on the digestion of food molecules.

(5)

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(Total for Question 7 = 11 marks)

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2. Question 7 · part c · 5 marks · Working and answer

Working and answer

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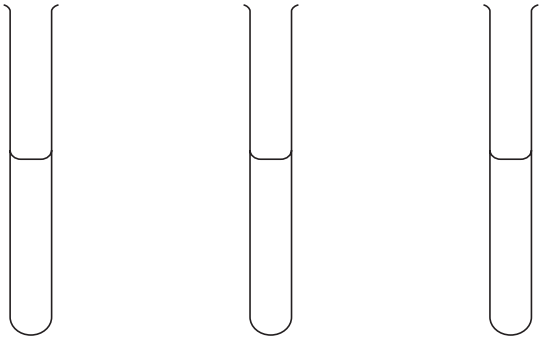
3. Question 2 · part a.iii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.iii only.

2 A student investigates the effect of bile salts on the digestion of lipids. The student uses milk as a source of lipids.

The student sets up three test tubes.



A **B** **C**

Table 1 lists the contents of each test tube.

Test tube A	Test tube B	Test tube C
5 cm^3 of milk	5 cm^3 of milk	5 cm^3 of milk
2 cm^3 of sodium hydrogencarbonate	2 cm^3 of sodium hydrogencarbonate	2 cm^3 of sodium hydrogencarbonate
6 drops of phenolphthalein	6 drops of phenolphthalein	6 drops of phenolphthalein
1 cm^3 enzyme	1 cm^3 enzyme	1 cm^3 boiled enzyme
distilled water	bile salts	bile salts

Table 1

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3. Question 2 · part a.iii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.iii only.

The student records the colour of the contents of each tube at the start of the investigation, and at 5 minute intervals for 15 minutes.

Table 2 shows the student's results.

	Colour of contents		
	Tube A	**Tube B**	**Tube C**
at start	pink	pink	pink
after 5 minutes	pink	colourless	pink
after 10 minutes	pink	colourless	pink
after 15 minutes	colourless	colourless	pink

Table 2

Phenolphthalein is pink in solutions above pH 10 and colourless in solutions below pH 8.

(a) (i) Which enzyme is used in this investigation? **(1)**

- ☒ **A** amylase
- ☒ **B** carbohydrase
- ☒ **C** lipase
- ☒ **D** protease

(ii) Which chemical elements are found in lipids? **(1)**

- ☒ **A** carbon, hydrogen and oxygen
- ☒ **B** carbon, hydrogen, oxygen and nitrogen
- ☒ **C** carbon, hydrogen, oxygen and sulfur
- ☒ **D** carbon, hydrogen, oxygen, nitrogen and sulfur

(iii) Where in the body are bile salts produced? **(1)**

- ☒ **A** brain
- ☒ **B** duodenum
- ☒ **C** liver
- ☒ **D** pancreas

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3. Question 2 · part a.iii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.iii only.

(b) Explain the purpose of tube C.

(2)

(c) Suggest one reason for the addition of the sodium hydrogencarbonate.

(2)

(d) Describe the effect of bile salts on lipid digestion in this investigation.

(3)

(Total for Question 2 = 10 marks)

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3. Question 2 · part a.iii · 1 mark · Working and answer

Working and answer

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4. Question 2 · part d · 3 marks

November 2020 Paper 2

Further practice · Attempt part d only.

2 A student investigates the effect of bile salts on the digestion of lipids. The student uses milk as a source of lipids.

The student sets up three test tubes.

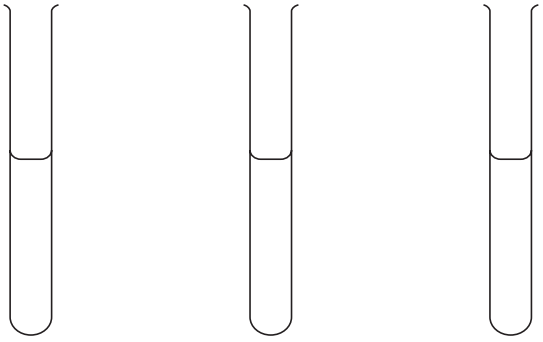


Table 1 lists the contents of each test tube.

Test tube A	Test tube B	Test tube C
5 cm^3 of milk	5 cm^3 of milk	5 cm^3 of milk
2 cm^3 of sodium hydrogencarbonate	2 cm^3 of sodium hydrogencarbonate	2 cm^3 of sodium hydrogencarbonate
6 drops of phenolphthalein	6 drops of phenolphthalein	6 drops of phenolphthalein
1 cm^3 enzyme	1 cm^3 enzyme	1 cm^3 boiled enzyme
distilled water	bile salts	bile salts

Table 1

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4. Question 2 · part d · 3 marks

November 2020 Paper 2

Further practice · Attempt part d only.

The student records the colour of the contents of each tube at the start of the investigation, and at 5 minute intervals for 15 minutes.

Table 2 shows the student's results.

	Colour of contents		
	Tube A	**Tube B**	**Tube C**
at start	pink	pink	pink
after 5 minutes	pink	colourless	pink
after 10 minutes	pink	colourless	pink
after 15 minutes	colourless	colourless	pink

Table 2

Phenolphthalein is pink in solutions above pH 10 and colourless in solutions below pH 8.

(a) (i) Which enzyme is used in this investigation?

(1)

- ☒ **A** amylase
- ☒ **B** carbohydrase
- ☒ **C** lipase
- ☒ **D** protease

(ii) Which chemical elements are found in lipids?

(1)

- ☒ **A** carbon, hydrogen and oxygen
- ☒ **B** carbon, hydrogen, oxygen and nitrogen
- ☒ **C** carbon, hydrogen, oxygen and sulfur
- ☒ **D** carbon, hydrogen, oxygen, nitrogen and sulfur

(iii) Where in the body are bile salts produced?

(1)

- ☒ **A** brain
- ☒ **B** duodenum
- ☒ **C** liver
- ☒ **D** pancreas

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4. Question 2 · part d · 3 marks

November 2020 Paper 2

Further practice · Attempt part d only.

(b) Explain the purpose of tube C.

(2)

(c) Suggest one reason for the addition of the sodium hydrogencarbonate.

(2)

(d) Describe the effect of bile salts on lipid digestion in this investigation.

(3)

(Total for Question 2 = 10 marks)

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4. Question 2 · part d · 3 marks · Working and answer

Working and answer

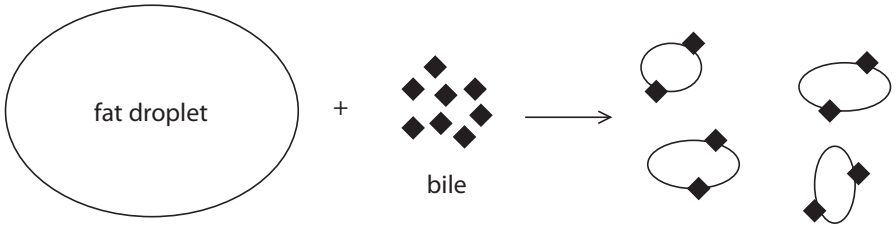
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5. Question 3 · part a.i · 1 mark

January 2023 Paper 1

Further practice · Attempt part a.i only.

3 (a) The diagram represents the effect of bile on a large fat droplet.



(i) Where is bile produced?

(1)

- ☒ **A** gall bladder
- ☒ **B** liver
- ☒ **C** small intestine
- ☒ **D** stomach

(ii) Describe the role of bile in the digestion of fat.

(3)

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5. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

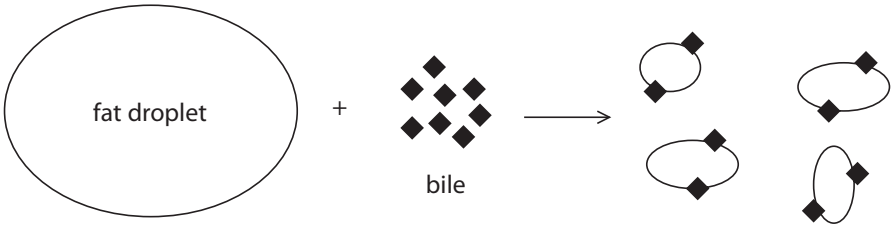
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6. Question 3 · part a.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part a.ii only.

3 (a) The diagram represents the effect of bile on a large fat droplet.



(i) Where is bile produced?

(1)

- ☒ **A** gall bladder
- ☒ **B** liver
- ☒ **C** small intestine
- ☒ **D** stomach

(ii) Describe the role of bile in the digestion of fat.

(3)

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6. Question 3 · part a.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part a.ii only.

(b) Three food tests are used on a solution of food.

The table shows the reagent used and the colour of the food solution after each test.

Reagent used	Colour of the food solution after the test
iodine solution	blue/black
biuret solution	purple
Benedict's solution	blue

(i) Describe what the food tests show about the dietary components in the food.

(3)

(ii) Saliva is added to a new sample of the food solution and the mixture is kept at 37 °C for 10 minutes.

The food tests are then repeated and the solution turns brick red when tested with Benedict's reagent.

Explain the effect of saliva on the food solution.

(3)

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6. Question 3 · part a.ii · 3 marks · Working and answer

Working and answer

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7. Question 7 · part c · 3 marks

November 2023 Paper 2

Further practice · Attempt part c only.

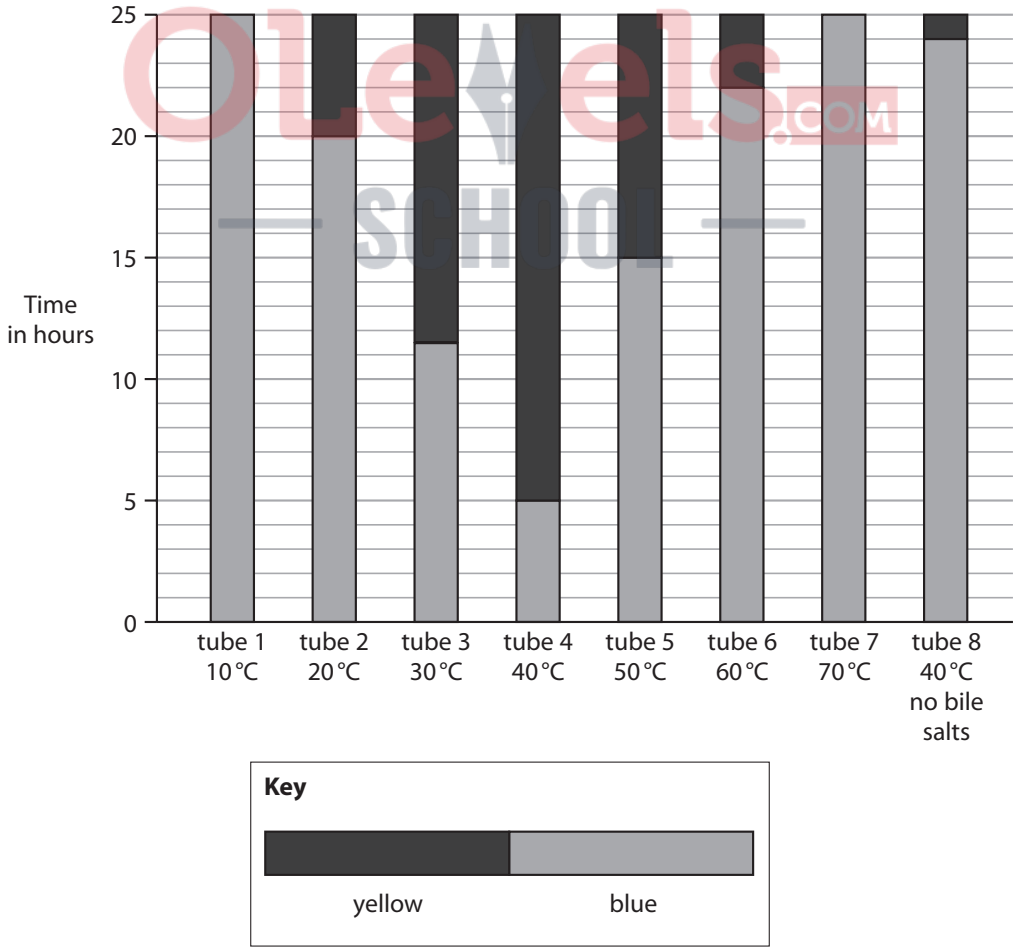
7 A student investigates the digestion of olive oil (lipid) using the enzyme lipase, bile salts and a blue indicator.

The indicator changes from blue to yellow when the pH is below 5.

This is the student's method.

- mix together olive oil, lipase solution, bile salts and one drop of indicator
- put this mixture into a test tube
- repeat the process for six other test tubes
- keep the tubes at different temperatures
- keep another test tube (test tube 8) containing just olive oil, indicator and lipase at 40 °C
- record the colours of the mixture of each tube every hour for 25 hours

The chart shows the student's results.



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7. Question 7 · part c · 3 marks

November 2023 Paper 2

Further practice · Attempt part c only.

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(a) (i) State three factors that should be kept constant in the student's investigation.

(3)

1 .....

2 .....

3 .....

(ii) State the optimum temperature for the action of lipase.

(1)

.....

(iii) Discuss why this temperature is only an approximate optimum temperature.

(3)

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(b) (i) Determine how much more time the indicator takes to change colour in tube 8 than in tube 3.

(2)

time = ... hours

(ii) Explain why the pH decreases during the 25 hours.

(3)

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7. Question 7 · part c · 3 marks

November 2023 Paper 2

Further practice · Attempt part c only.

(c) Explain the result for tube 8.

(3)

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(Total for Question 7 = 15 marks)

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Working and answer

8. Question 3 · part b · 2 marks

June 2024 Paper 2

Further practice · Attempt part b only.

3 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

People make lifestyle choices that can sometimes have a negative effect on their health. People who drink too much alcohol are at risk of brain and liver damage, certain types of cancer and mental illness. Liver damage can affect the digestion of fat. Alcohol can also affect the eyes, increasing the risk of developing cataracts that cause the lenses in the eyes to become cloudy. Alcohol is also a depressant. It increases the activity of a chemical in parts of the brain that control balance, speech, motor skills and memory. In addition, alcohol suppresses an area of the brain called the amygdala. The amygdala is responsible for the fight or flight function which prepares the body for action, initiating the release of stress hormones. These hormones increase glucose usage by the brain and other parts of the body.

When a person stops drinking alcohol after years of drinking a lot of alcohol, the amygdala in the brain becomes fully active again, which is the cause of all forms of alcohol withdrawal. Over time, if a person regularly drinks heavily, the central nervous system gets used to the suppressing effect of the alcohol, which means the brain and body are affected if the alcohol level suddenly drops. An individual can go straight into fight or flight mode as the alcohol leaves the system and levels of stress hormones begin to return to normal. Symptoms of withdrawal can be as severe as seizures and hallucinations or less severe symptoms such as loss of appetite and insomnia (an inability to sleep).

(a) Drinking alcohol and diet are two lifestyle choices.

Describe how a lifestyle choice, other than drinking alcohol and diet, can have a harmful effect on health (Lines 1 and 2).

(2)

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8. Question 3 · part b · 2 marks

June 2024 Paper 2

Further practice · Attempt part b only.

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(b) Explain how drinking alcohol affects the digestion of fat (Lines 3 to 4).

(2)

(c) State how cataracts affect vision (Lines 4 to 5).

(1)

(d) Name a part of the brain, other than the amygdala, that is affected by alcohol (Lines 5 to 7).

(1)

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(e) Explain how alcohol acts as a depressant on the nervous system (Line 5).

(3)

8. Question 3 · part b · 2 marks · Working and answer

Working and answer

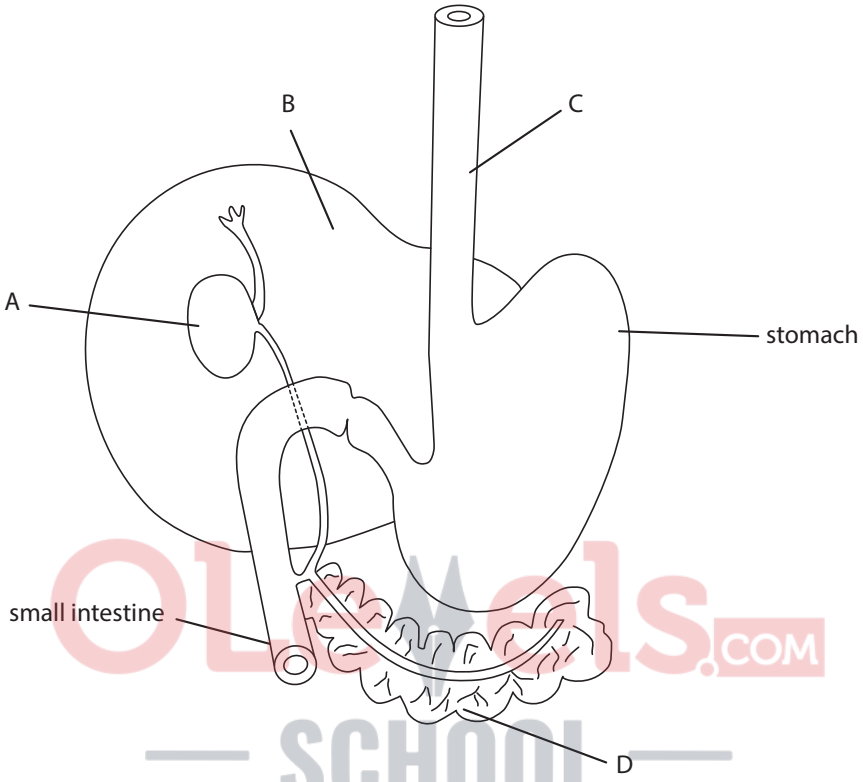
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9. Question 2 · part a.ii · 1 mark

June 2025 Paper 2

Further practice · Attempt part a.ii only.

2 The diagram shows structures involved in digestion.



(a) (i) Identify the structures labelled A, B, C and D.

(4)

A .....

B .....

C .....

D .....

(ii) Name the fluid stored in structure A.

(1)

.....

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9. Question 2 · part a.ii · 1 mark

June 2025 Paper 2

Further practice · Attempt part a.ii only.

(iii) Explain the function of the fluid stored in structure A in digestion.

(3)

(b) (i) Describe the role of structure D in digestion.

(4)

(ii) Name two substances secreted by structure D that are not involved in digestion.

(2)

1 .....

2 .....

(Total for Question 2 = 14 marks)

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9. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

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